

# TEnet: target speaker extraction network with accumulated speaker embedding for automatic speech recognition

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It is challenging to perform automatic speech recognition when multiple people talk simultaneously. To solve this problem, speaker-aware selective methods have been proposed to extract the speech of the target speaker, relying on the auxiliary speaker characteristics provided by an anchor (a clean audio sample of the target speaker). However, the extraction performance depends on the duration and quality of the anchors, which is unstable. To address this limitation, the authors propose a target speaker extraction network (TEnet) which applies the robust speaker embedding to extract the target speech from the speech mixture. To get more stable speaker characteristics during training, the robust speaker embeddings are accumulated over all the speech of each target speaker, rather than utilising the embedding produced by a single anchor. As for testing, very few anchors are enough to get decent extraction performance. Results show the TEnet trained with accumulated embedding achieves better performance and robustness compared with the single-anchored TEnet. Moreover, to exploit the potential of the speaker embedding, the authors propose to feed the extracted target speech as anchor and train a feedback TEnet, whose results are superior to the short-anchored baseline for 22.5% on word error rate and 15.5% on signal-to-distortion rate.

**Introduction:** Multi-talker speech recognition is a long-lasting problem in automatic speech recognition (ASR), and the multi-talker scenarios are quite common in many practical applications, including mobile voice assistants, smart home devices, audio content monitoring. Recently, many deep learning based approaches have been explored to solve this problem, which falls into blind source separation methods and speaker aware-speech extraction ways.

Deep neural network (DNN) is often trained to estimate the T-F mask for clean signal from a mixed corrupted signal [1]. In source separation methods like Deep clustering [2–4], Deep attractor network [5] and Tasnet [6], the neural networks take the mixed speech (for  $n$  speakers) as input and produce  $n$  output masks corresponding to  $n$  speakers. To solve the arbitrary source permutation problem, the permutation invariant training (PIT) [7–10] has been proposed, in which the different sources are treated as a whole set and the network is trained to minimise the assignment with minimum error.

As for speaker-aware extraction, a clean register audio sample of the target speaker (e.g. assistant devices user) is often provided as the anchor. From the anchor, we can obtain the speaker characteristic for the target speaker and enhance the target speech extraction [11]. Related work such as DEnet [12], Speakerbeam [13], and Voicefilter [14] show promising performances. These speaker aware extraction networks take the mixed speech and auxiliary speaker characteristics (from anchors) to produce the speech for the target speaker in both training and testing stages.

In the recent speaker-aware speech extraction ways, a single random chosen anchor is often used to produce the speaker characteristics and enhance the extraction ability. Since the voice of the target speaker can be varying with different contents and situations, the characteristics of the same speaker in the random anchor and mixed speech might be different. This will lead to an unstable extraction result. Besides the anchor is very short sometimes and we find that the duration and quality of the anchor have a big impact on the performance of the target speaker extraction network (TEnet). To address this limitation, the proposed TEnet adopts the robust speaker embedding that is generated by a speaker recognition network with an aggregation layer. To get more stable long-term speaker characteristics during the training of the TEnet, we accumulate the robust speaker embeddings and average them over all the speech of each target speaker, rather than utilising the speaker embedding produced by a single anchor. While testing, we just use single or very few anchors, which makes it easy to generalise over unseen speakers. To further exploit the potential of more accurate speaker embedding, we propose to feed the extracted target speech as the anchor and train a feedback TEnet, which benefits from the accurate speaker information and achieves higher extraction quality.

**The robust speaker embedding network:** Using DNN to extract speaker characteristics is a popular research area [15–19], in which a speaker

recognition DNN is trained, with speaker embeddings extracted from the middle of the DNN. Here we apply a robust DNN-based speaker embedding called x-vector [18] to guide the target speech extraction in the TEnet. To get the x-vector, a speaker embedding network should be trained to distinguish different speakers. In the middle of the network, an aggregation layer collects the statistics of the previous frame-level layers, computing the mean and standard deviation over input segment (e.g. 300 frames). Then, the mean and standard deviation are concatenated and passed to the following segment-level layers. The segment-level layers can be used as the speaker embedding layers. Finally, a softmax output layer is added for speaker classification. The output dimension of this model is equal to the number of speakers in the training data. After training, the segment-level layers are able to generate distinguishing speaker embeddings and map the utterance with variable length into fixed-dimensional embeddings. The purpose of this network is to extract robust speaker embeddings which can generalise well to unseen speakers.

**The target speaker extraction network:** The aim of TEnet is to extract the speech of the target speaker from the speech mixture. It takes the robust speaker embedding for the target speaker and the mixed speech to produce the spectrum of the target speaker. The study in paper [20] indicates that attention plays an important role in the selective enhancement in the cocktail party problem. The enhancement is getting easier on the signal with triggered attention. In the TEnet, the accumulated robust speaker embeddings that encode the characteristics for the target speaker can be treated as the attention trigger, which will help gathering the T-F bin of the to-be-selected speaker.

Instead of using a single speech as the anchor, we propose to accumulate the robust speaker embeddings over all the speech of the same speaker to train the TEnet. It exploits the stable speaker characteristic for each target speaker to enhance the extraction ability during training. As for testing, we just provide single or very few anchors to extract the speaker embeddings. By this means, the training procedure of TEnet learns much constructability benefiting from the stable accumulated speaker embeddings. And with just a few anchors for testing, the model is easy to generalise over unseen speakers.

The framework of the TEnet is shown in Fig. 1. To train this model, the multi-talker mixed speech, the individual clean target speech in the mixture (reference clean speech) and the averaging accumulated speaker embedding are required. The TDNN in Fig. 1 is the speaker embedding model. The TEnet consists of two convolutional neural network (CNN) layers, three bi-direction long short-time memory (BLSTM) layers and one full-connected (FC) layer. CNN layers take the spectrum of mixed speech as input. The following BLSTM layers accept the output of the CNN layers and the accumulated speaker embedding. We only apply convolution operation on the speech spectrum, because the speaker embedding is already a compact representation. After the BLSTM layers, the FC layer maps the output dimension equal to the signal spectrum, which is 257.

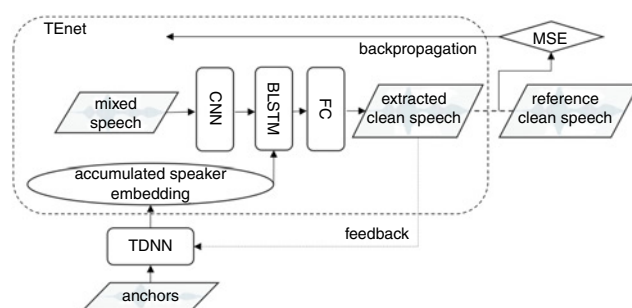


Fig. 1 The architecture of the proposed TEnet

The TEnet takes the audio pair and speaker embedding to produce the phase sensitive mask (PSM) [1] of the target speaker. The PSM is defined on Short-time Fourier transform between the clean speech and the mixed speech pair as follows:

$$M_{\text{PSM}}(t, f) = \frac{S(t, f)}{Y(t, f)} \quad (1)$$

where  $|S(t, f)|$  and  $|Y(t, f)|$  indicate the spectral magnitude of the clean and mixed speech respectively. The TEnet is trained to estimate the

target speech's PSM, which can be applied to recover the spectrum of target speech by element-wise multiplication with mixed speech. The loss function is the mean square error (MSE) between the spectrum of the reference clean target speech and the masked one, which minimises the following objective function

$$\mathcal{L}_{\text{MSE}} = \sum_{t,f} \|S_{t,f} - Y_{t,f} \times M_{t,f}\|_2^2 \quad (2)$$

where  $S$  is the reference spectrum (time  $t$  and frequency  $f$ ),  $Y$  represents the speech mixture, and  $M$  indicates the estimated mask.

In the inference stage, only the mixed speech and the anchors for the target speaker are needed. If more than one anchor is provided, simply averaging the speaker embeddings of several anchors (e.g. three) will benefit the extraction result a lot.

To further improve the accuracy of the speaker embeddings, we propose to utilise the resulted extracted speech by TEnet, treating it as a feedback anchor to generate the speaker embedding and train a feedback TEnet. The information flow is shown in Fig. 1 as the dashed line named 'feedback'. With the predicted target speech, the speaker embeddings are able to capture more accurate speaker information for the current speech. Moreover, some additional cues such as speech content might strengthen the attention trigger. The parameters of this feedback TEnet are initialised with a reference-anchored model, in which the reference clean speech for the target speaker is applied as both anchor and training label since the reference speech contains the most accurate speaker information guiding the TEnet. During training, we use the average embedding TEnet to extract the target speech, which is used as the anchor to train this feedback TEnet. In the testing stage, the feedback anchor should also be produced by the TEnet in the first pass.

**Data set and model configurations:** The speaker embedding model was trained independently using Kaldi [21] toolkit. The training data contained part of data from Switchboard data set and the NIST speaker recognition evaluation data. Moreover, the data augmentation technique [19] was applied to increase the diversity of the training data. To ensure enough data for each speaker in the training data, the speakers with fewer than eight utterances were removed. Finally, there were totally 5195 speakers left, meaning that the output dimension for the model was 5195. The feature was 23D MFCC. The speaker embedding network consisted of a seven-layer 512-node TDNN [22] and a 1500-node statistic aggregation layer after the fifth TDNN layer. Suppose the input segment contains  $T$  frames (around 300 frames), the first five layers made a computation on the speech frames. Then the statistic aggregation layer (1500-node) collected the outputs of the fifth layer over  $T$  frames in the segment, calculating their mean and standard deviation. The mean and deviation were appended to form the 3000D input to the last two layers. After the speaker discriminative training, the robust embeddings can be extracted from the sixth TDNN layer, which was 512D compact speaker embeddings for the target speaker.

As for the TEnet, we used part of data from *Wall Street Journal* (WSJ) corpus [23], generating the two-talker mixture (WSJ0-2mix) data set. This data set has already been applied in many works for speech separation [2, 3, 7]. The details of this data set were mentioned in paper [2]. The training and validation data consisted of 30 and 10 h mixed speech for 101 speakers, and the test set included 5 h speech mixture for 16 unseen speakers. To generate the mixed speech, we randomly chose an interfering speech from another speaker and then mixed it with the target clean speech. It was done by summing the clean target speech and interfering speech with different signal-to-noise ratios between 0 and 5 dB.

The input of TEnet was the 257D log spectrum. We used 2-layer CNN, 3-layer 640-node BLSTM, and 1-layer FC. The kernel sizes were  $9 \times 9$  and the output channels were 64 for both CNN layers. Then the FC layer accepted the output of the last BLSTM layer, generating the 257D mask for the target speech. This mask was multiplied with the input mixed spectrum to obtain the clean spectrum, which was the final output of the TEnet. TensorFlow [24] was used to build the TEnet.

**Experiments and results:** As shown in Table 1, the third and fourth columns show the anchor types used for training and testing respectively

- 'short': the anchor applied to generated speaker embedding is a random utterance shorter than 3 seconds;

**Table 1:** WER (%) and SDR (dB) for TEnet with different anchors

| num | Model            | Anchor   |            | WER          | SDR          |
|-----|------------------|----------|------------|--------------|--------------|
|     |                  | Train    | Test       |              |              |
| 1   | uPIT-blstm [8]   | —        | —          | —            | 9.4          |
| 2   | TASnet [6]       | —        | —          | —            | 11.1         |
| 3   | PIT-e2e [10]     | —        | —          | 25.4         | —            |
| 4   | Speakerbeam [13] | long     | long       | 34.0         | —            |
| 5   | uPIT-cnnblstm    | —        | —          | 33.06        | 9.7          |
| 6   | Voicefilter      | long     | long       | 32.82        | 10.02        |
| 7   | —                | —        | —          | 78.22        | 1.12         |
| 8   | TEnet            | short    | short      | 34.69        | 9.78         |
| 9   | TEnet            | long     | long       | 31.99        | 10.17        |
| 10  | TEnet            | per-spk  | short      | 32.55        | 10.21        |
| 11  | TEnet            | per-spk  | long       | 31.0         | 10.37        |
| 12  | TEnet            | per-spk  | 3 average  | 29.56        | 10.93        |
| 13  | TEnet            | per-spk  | 5 average  | <b>29.2</b>  | <b>11.0</b>  |
| 14  | TEnet            | per-spk  | 20 average | 29.17        | 10.99        |
| 15  | TEnet            | per-spk  | per-spk    | 29.06        | 10.98        |
| 16  | TEnet            | feedback | feedback   | <b>26.89</b> | <b>11.30</b> |
| 17  | +AM fine-tune    | feedback | feedback   | <b>11.60</b> | <b>11.30</b> |

- 'long': the anchor is a random utterance around 10 seconds;
- 'per-spk': we average the speaker embeddings over all the utterance of the same speaker in the whole training data set;
- 'n-average': we randomly choose  $n$  utterance for the target speaker to compute the average speaker embedding;
- 'feedback': the feedback mechanism is applied.

The chosen anchors were different from the clean speech for the target speakers in the current speech mixture. The performance of the TEnet was evaluated in terms of two measurements: word error rate (WER) for speech recognition and signal-to-distortion rate (SDR) [25] for signal quality. The ASR acoustic model (AM) for speech recognition was a five-layer 650-node TDNN model, trained with the cross-entropy criteria following the standard setup in Kaldi WSJ example. We used the WSJ1 SI284 to train the AM.

The first block in Table 1 shows some state-of-art published results for related works on the WSJ0-2mix. We also trained an utterance-level PIT model [8] (row 5) and a Voicefilter [14] network (row 6) using the same data and model configurations with the TEnet. Row 7 presents the WER and SDR for the mixed speech. The results of the proposed TEnet are presented on the following rows. Rows 8 and 9 are two baselines using short and long anchors for both training and testing stages, which indicates that the duration of the anchor has a big impact on the extraction performance for the single-anchor TEnet. With average accumulated speaker embeddings during training, the TEnet achieves better performance on both WER and SDR, as shown in rows 10 and 11. It improves the constructability of TEnet, especially when the anchor is short. And training the TEnet with average  $x$ -vector, the gap between using a short and long anchor for testing becomes much smaller. During testing, the TEnet can take advantage of averaging the speaker embeddings on several anchors for unseen speakers (rows 12–15). To balance the extraction performance and the number of the anchors needed for testing, we prefer five anchors for the target speakers to generate average accumulated speaker embeddings. It is easy to get three to five anchors of the target speakers, which makes it practical and effective to do extraction with average speaker embeddings. It achieves 15.8 and 12.5% relative improvements on the WER and SDR compared with the short anchor method. Comparing with other related works, we find that the robust speaker embedding is helpful to enhance the speech extraction performance.

Then we apply the TEnet to extract the target speech of the training data and feed the extracted target speech as the anchor to train a feedback TEnet. During testing, we use the per-spk TEnet (with five average testing anchor) to obtain the feedback anchor. The results are shown in row 16, whose promotion is about 22.5% on WER and 15.5% on SDR compared with the baseline. From this, we can see that the more accurate speaker information and the additional cues provided by the feed-back anchor helps with better extraction ability. It further proves that the accuracy of the provided speaker information is important for the TEnet. Then we apply the TEnet to do the extraction on the mixed training data set. And the resulted extracted audios with

WER lower than 30% (decoding by the clean AM) are selected to train a fine-tuning AM. The parameters of the fine-tuning AM are initialised with the clean AM. The results of the fine-tuning AM is shown in the last row in Table 1, which is a large promotion on the recognition accuracy.

**Conclusions:** In this paper, the speaker-aware TEnet is proposed to extract speech for the target speaker from the speech mixture. We find that the quality and robustness of the speaker embedding play important roles in the TEnet. Thus we decided to apply the robust speaker embedding to enhance the extraction ability. To get more stable speaker information, we improve the recent single-anchor-based extraction with the average accumulated speaker embeddings. During the training of the TEnet, the speaker embeddings are accumulated and averaged on the whole training data set to exploit stable long-term speaker characteristics for the target speakers. On the testing stage, very few anchors are enough to achieve decent extraction results, which makes it a practical and effective way to use TEnet with accumulated speaker embeddings. Moreover, to exploit the potential of more accurate speaker embedding, the feedback mechanism is proposed to further improve the extraction capacity of the TEnet. The experiment results show that the proposed TEnet is able to outperform the single short-anchor baseline for about 22.5% on WER and 15.5% on SDR relatively. Comparing with other related works, the TEnet performs better as well.

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One or more of the Figures in this Letter are available in colour online.

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